

# AECCS: Assessing the Effectiveness of Cookie Consent Systems

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CS475 Privacy-Enhancing Technologies · Final report template aligned with the current repository state

## Abstract

AECCS is an implementation-focused measurement system for studying cookie consent systems, GDPR-oriented compliance signals, dark patterns, and privacy-enhancing technologies (PETs). The repository now contains a provenance-aware pipeline that separates synthetic development data from the final real-study dataset. The current implementation supports automated crawling, tracker classification, dark-pattern detection, compliance scoring, browser PET evaluation, CMP comparison, differential privacy reporting, and report generation. The remaining gap is empirical rather than architectural: the real crawl and real PET experiments must be executed under the `data/real/` tree before final claims are made.

The metrics presented below are from the completed 100-site real study executed on 1 March 2026 under run ID `real-study-20260301-final`, with EU geolocation via Proton VPN (Netherlands). The synthetic dataset in `data/mock/` remains available for pipeline validation.

## 1. Problem Statement

Cookie consent systems are intended to provide meaningful user control over non-essential tracking. Prior work and regulatory guidance suggest that many banners either manipulate users toward acceptance or fail to technically enforce rejection. The core question of this project is whether consent systems protect privacy in practice and, when they do not, whether PETs reduce user exposure.

## 2. Related Work

AECCS is grounded in three main strands of prior work: studies of dark patterns in consent banners, studies of enforcement failures after rejection, and large-scale automated cookie-consent measurement. A companion literature review is included in `docs/literature_review.md`.

## Key references

- Nouwens et al., *Dark Patterns after the GDPR*
- Matte et al., *Do Cookie Banners Respect My Choice?*
- Bollinger et al., *Automating Cookie Consent and GDPR Violation Detection*
- GDPR Regulation (EU) 2016/679
- CNIL cookie and tracker guidance

## 3. System Overview

The implemented pipeline follows the proposal closely. Each target site is visited in three states: no interaction, accept all, and reject all. Raw crawl outputs are then processed into tracker classifications, dark-pattern findings, compliance scores, aggregate statistics, PET comparisons, and report artifacts.

| Module                           | Status                          |
|----------------------------------|---------------------------------|
| Crawler                          | Implemented                     |
| Tracker classification           | Implemented                     |
| Dark-pattern detection           | Implemented                     |
| Compliance scoring               | Implemented                     |
| Aggregate metrics                | Implemented                     |
| Browser PET evaluation           | Implemented, awaiting real runs |
| CMP analysis                     | Implemented                     |
| Differential privacy reporting   | Implemented                     |
| Visualization and HTML reporting | Implemented                     |

## 4. Data Hygiene and Reproducibility

A major repository milestone in the current phase was separating historical mixed outputs from the active datasets. New runs now target `data/mock/` or `data/real/`, while older root-level artifacts have been quarantined under `data/legacy/` and `reporting/legacy/`.

Raw and processed artifacts now carry provenance fields including source mode, run ID, generation timestamp, browser information, proxy state, and study-list origin. The report generator rejects mixed processed inputs to avoid blending mock and real findings.

### 5. Real-Study Findings (100 Sites)

The real study crawled 100 European and international websites on 1 March 2026 using Proton VPN (Netherlands) for EU geolocation. 97 sites were successfully crawled; 3 failed due to transient HTTP/2 or timeout errors. The PET evaluation covered 7 configurations across all 100 sites (700 measurements total, 681 successful).

| Metric                         | Real-study value                                |
|--------------------------------|-------------------------------------------------|
| Pre-consent tracker violations | 89.7%                                           |
| Sites with any dark pattern    | 82.0%                                           |
| Average compliance score       | 33.1 / 100 (median 24.0)                        |
| Best browser PET               | Brave Shields (14.7% average tracker reduction) |
| Best CMP                       | OneTrust (PET score 33.4)                       |
| Recommended DP epsilon         | 1.0                                             |

### 6. Proposal Coverage

The original proposal committed to a real-site automated measurement study over 100 to 200 websites. The repository now includes a 100-domain seed list in `data/websites.csv`, which brings the project back in line with the proposal at the planning level. What remains is execution of the final study under EU-like browsing conditions and regeneration of all figures and report outputs from real data only.

### 7. Completed Work

- Real crawl executed across 100 sites into `data/real/raw/` (97/100 successful).
- Full PET evaluation (7 PETs × 100 sites = 700 measurements) into `data/real/processed/`.
- All aggregate metrics, DP outputs, CMP comparison, 11 figures, and HTML report regenerated from real data.
- Demo findings in this document replaced with real-study findings.

### 8. Ethics and Limitations

The project uses automated browsing and therefore must remain rate-limited and transparent about its collection scope. The repository includes both synthetic data for pipeline validation and the completed real-study dataset from 100 sites. All findings reported above are based on the real-study run executed on 1 March 2026.

### 9. Conclusion

AECCS is now structurally ready for an honest final course submission. The core software system is built, the data model is provenance-aware, and the mock dataset validates the end-to-end pipeline. The final step is operational: execute the real measurement study and replace the demonstration findings with real evidence.

### References

Final ACM-format citations should be inserted here after the real-result draft is prepared. The literature review file already identifies the core sources to cite.